

MODULE - II

Beam

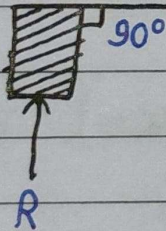
Beam

Beam is a structural member which has one dimension larger than other two dimensions, namely depth and width, and it is supported at few points.

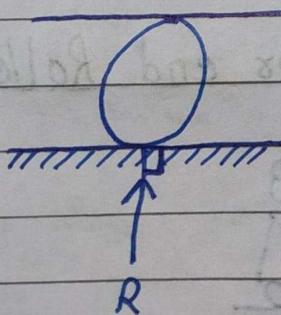
"The distance between two supports is called span."

Types of Support

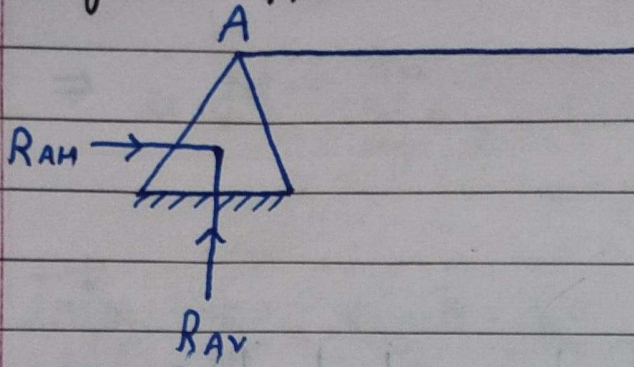
1. Simple Support:



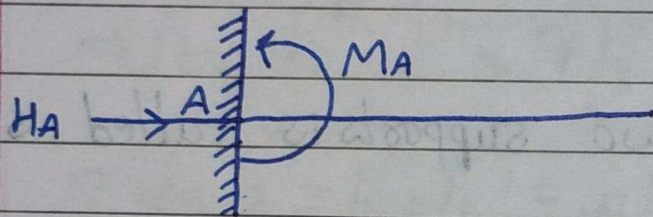
2. Roller Support:



3. Hinged Support:

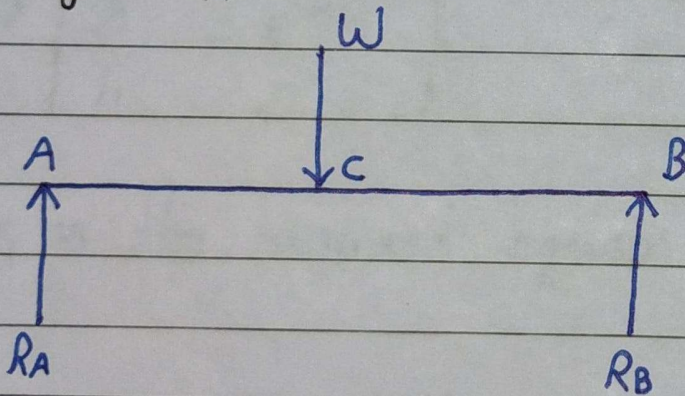


4. Fixed Support:

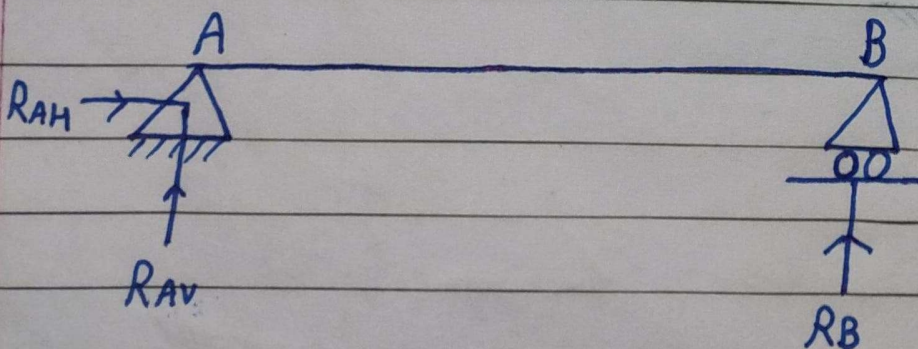


Types of Beam

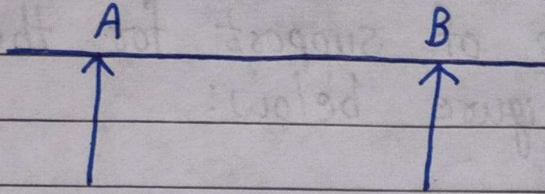
1. Simply Supported Beam:



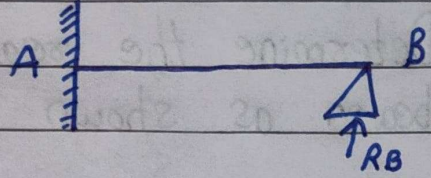
2. Beam with one end Hinged & other end Roller:



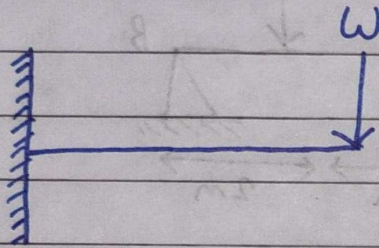
3. Over Hanged Beam:



6. Propped Cantilever Beam:

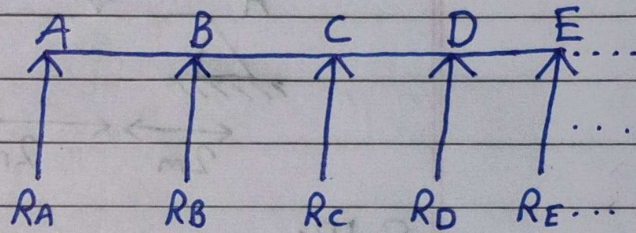


4. Cantilever Beam:

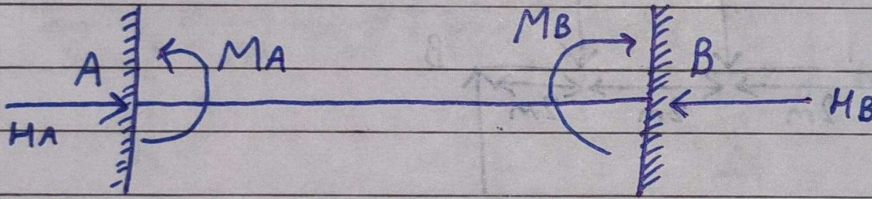


7. Continuous Beam:

[Supports more than two]

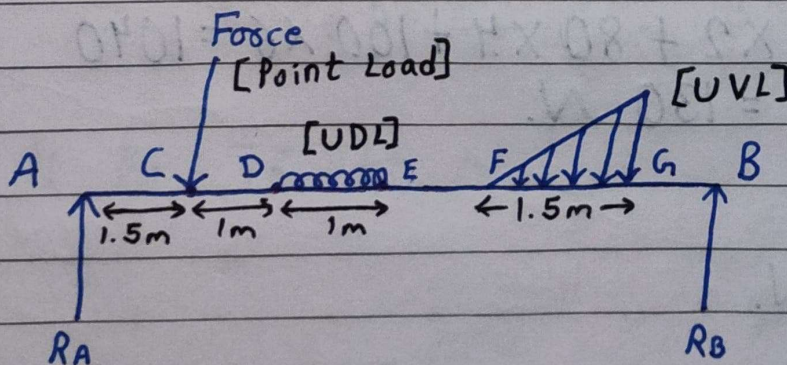


5. Fixed Beam:



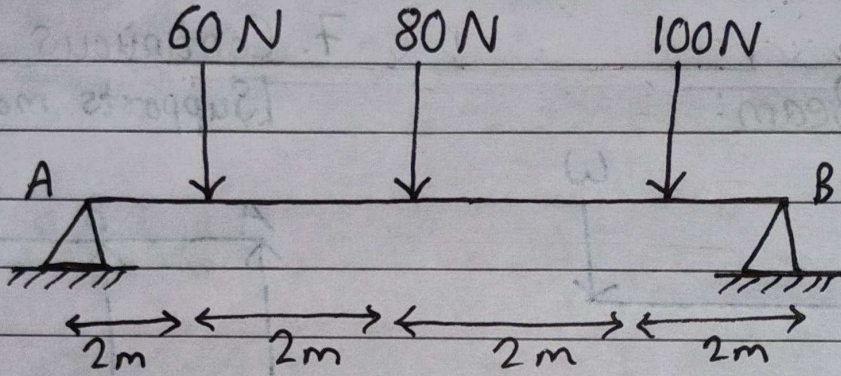
Types of Load

1. Point Load [C]
2. UDL (Uniformly Distributed Load) [D-E]
3. UVL (Uniformly Varying Load) [F-G]

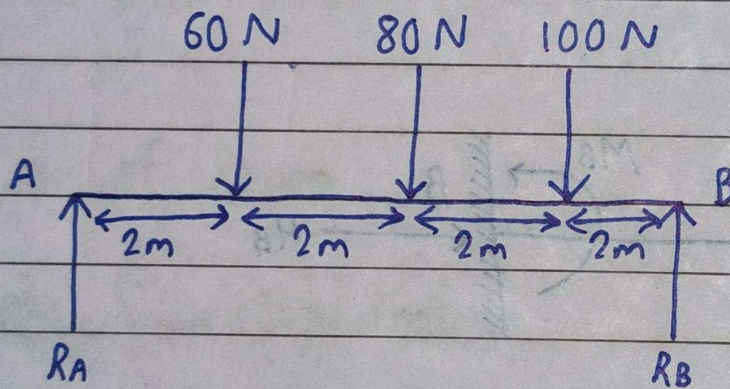


Numericals

Q) Determine the reactions of support for the following beam as shown in figure below:



Solⁿ:



$$\sum F_x = 0$$

$$\sum F_y = 0 \Rightarrow R_A + R_B = 60 + 80 + 100 = 240 \text{ N}$$

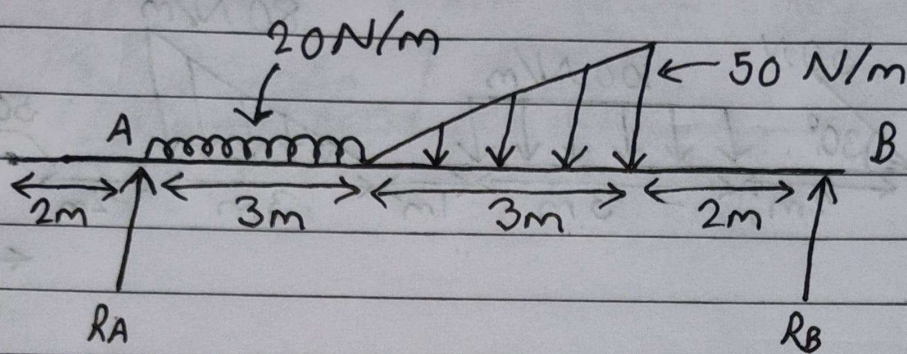
$$\sum M_A = 0$$

$$\Rightarrow R_B \times 8 = 60 \times 2 + 80 \times 4 + 100 \times 6 = 1040$$

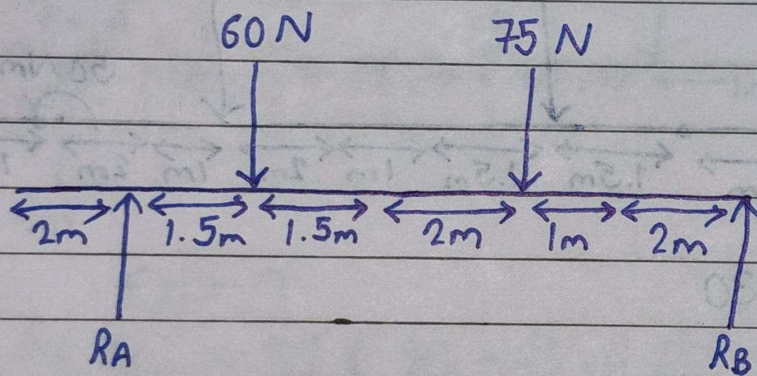
$$\Rightarrow R_B = \frac{1040}{8} = 130 \text{ N}$$

$$\therefore R_A = 110 \text{ N}$$

Q) Find the reactions at A & B for the following beam as shown in figure below:



Solⁿ:



$$\sum F_y = 0$$

$$\Rightarrow R_A + R_B = 60 + 75 = 135 \text{ N}$$

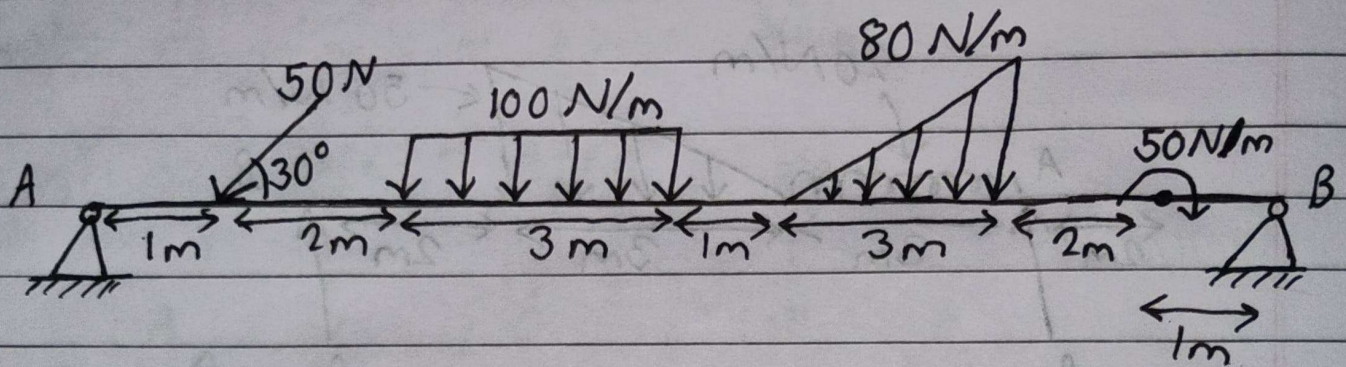
$$\sum M_A = 0$$

$$\Rightarrow R_B \times 8 = 60 \times 1.5 + 75 \times 5 = 465$$

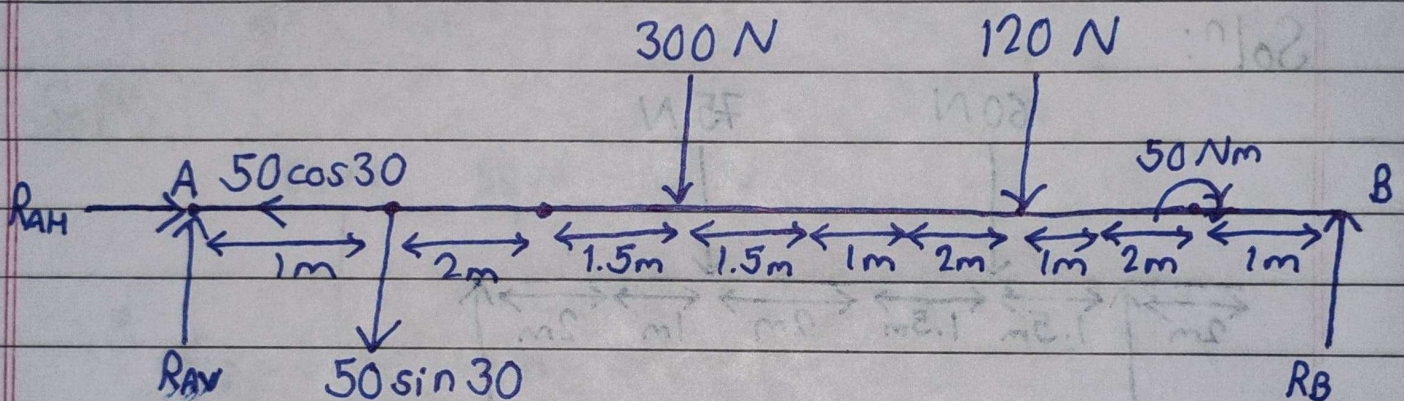
$$\therefore R_B = 58.125 \text{ N}$$

$$\& R_A = 76.875 \text{ N}$$

Q) Find the reactions at A & B for the following beam as shown in figure below.



Solⁿ:



$$\sum F_x = 0$$

$$\Rightarrow R_{AH} = 50 \cos 30 = 43.30 \text{ N.}$$

$$\sum M_A = 0$$

$$\Rightarrow R_B \times 13 = 1 \times 50 \sin 30 + 300 \times 4.5 + 120 \times 9 + 50$$

$$\Rightarrow R_B \times 13 = 2505$$

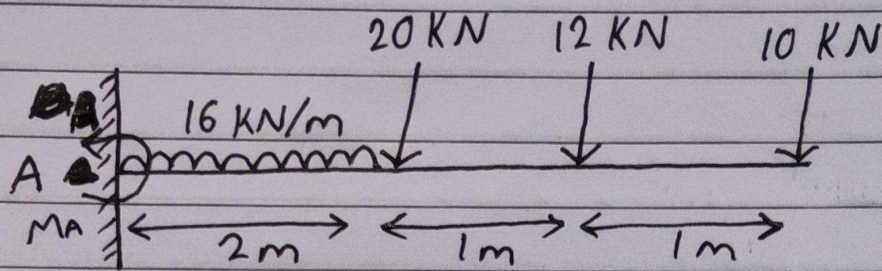
$$\therefore R_B = \frac{2505}{13} = 192.692 \text{ N.}$$

$$\sum F_y = 0 \Rightarrow R_{AV} + R_B = 50 \sin 30 + 300 + 120$$

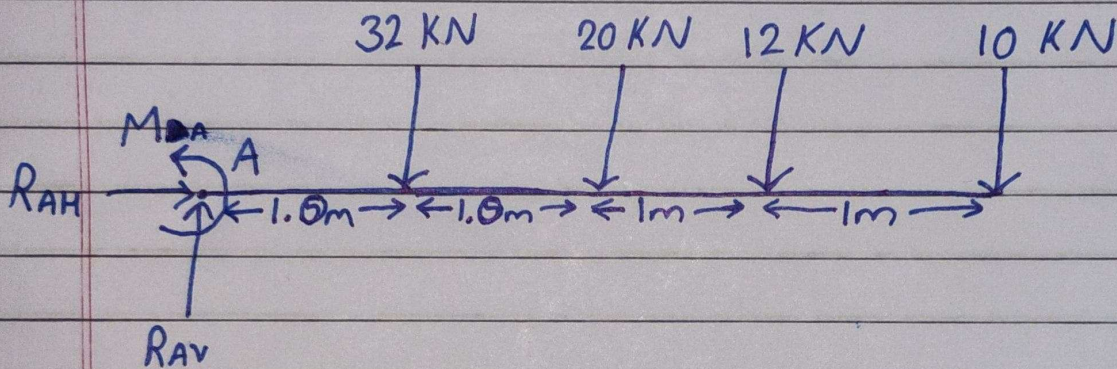
$$\Rightarrow R_{AV} + R_B = 445$$

$$\therefore R_{AV} = 252.308 \text{ N.}$$

Q) Determine the reactions for loaded cantilever beam as shown in figure below.



Solⁿ:



$$\sum Y = 0$$

$$\Rightarrow R_{AV} = 32 + 20 + 12 + 10 = 74 \text{ kN.}$$

$$\& R_{AH} = 0 \text{ N.}$$

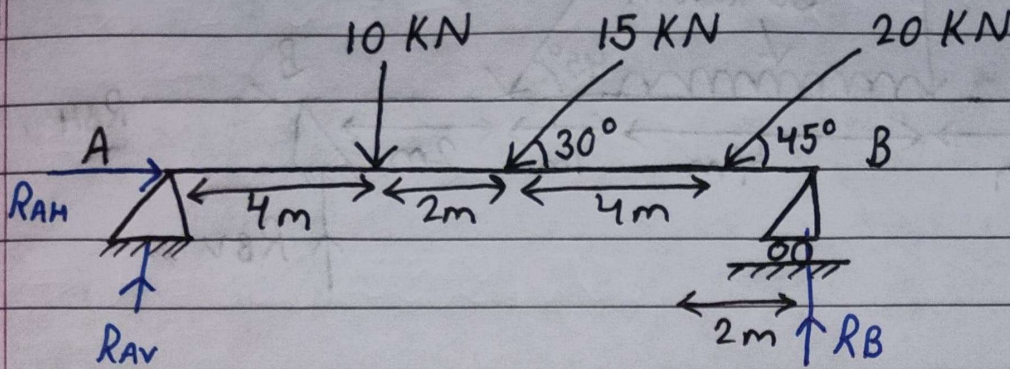
$$\text{Also, } \sum M_A = 0$$

$$\Rightarrow M_A = 32 \times 1 + 20 \times 2 + 12 \times 3 + 10 \times 4$$

$$\therefore M_A = 148 \text{ kNm.}$$

Numericals from Beam

Q)



Solⁿ:

$$R_{AH} = 15 \cos 30 + 20 \cos 45 = 27.132 \text{ kN}$$

$$R_{AV} + R_B = 10 + 15 \sin 30 + 20 \sin 45 = 31.642 \text{ kN}$$

$$\sum M_A = 0$$

$$\Rightarrow 10 \times 4 + 15 \times 6 \sin 30 + 20 \times 10 \sin 45 = R_B \times 12$$

$$\Rightarrow 185 = R_B \times 12 \quad \Rightarrow 226.421 = R_B \times 12$$

$$\Rightarrow R_B = 15.416 \text{ kN} \quad \Rightarrow R_B = 18.868 \text{ kN}$$

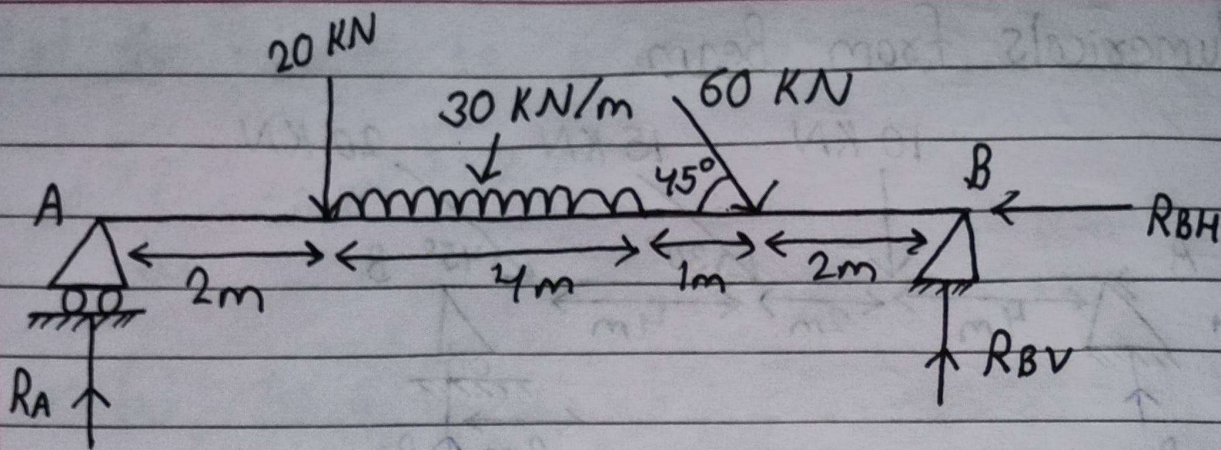
$$R_{AV} = 31.642 - 18.868 = 12.774 \text{ kN}$$

$$\therefore R_A = \sqrt{R_{AH}^2 + R_{AV}^2} = 29.988 \text{ kN}$$

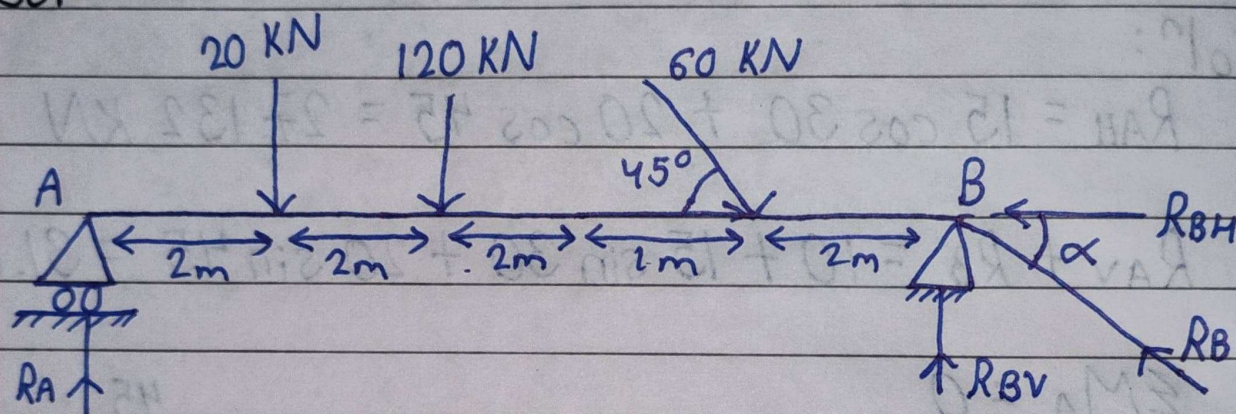
$$\tan \theta = \frac{R_{AV}}{R_{AH}}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{12.774}{27.132} \right) = 25.211^\circ$$

8)



Solⁿ:



$$R_{BH} = +60 \cos 45^\circ = +42.426 \text{ kN}$$

$$R_A + R_{Bv} = 20 + 120 + 60 \sin 45^\circ = 182.426 \text{ kN}$$

$$\sum M_A = 0$$

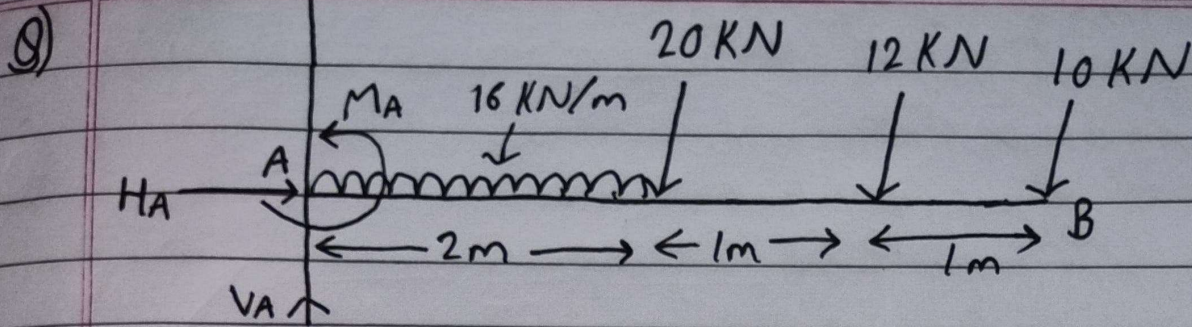
$$\Rightarrow 20 \times 2 + 120 \times 4 + 60 \times 7 \sin 45^\circ = R_{Bv} \times 9$$

$$\Rightarrow R_{Bv} = 90.776 \text{ kN}$$

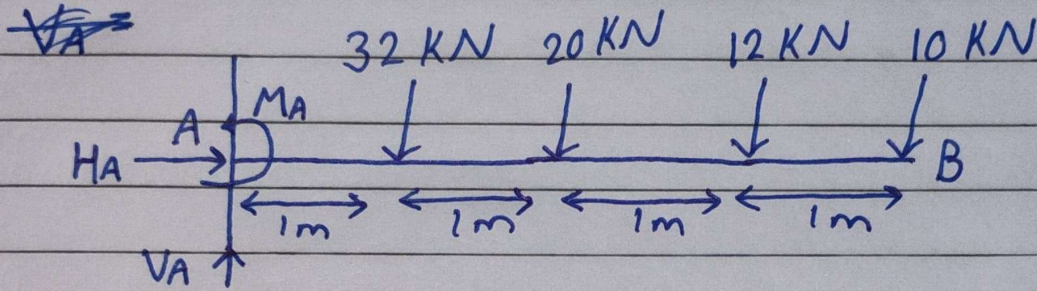
$$\therefore R_A = 91.649 \text{ kN}$$

$$R_B = \sqrt{R_{Bv}^2 + R_{BH}^2} = 100.201 \text{ kN}$$

$$\alpha = \tan^{-1} \left(\frac{R_{Bv}}{R_{BH}} \right) = 64.950^\circ$$



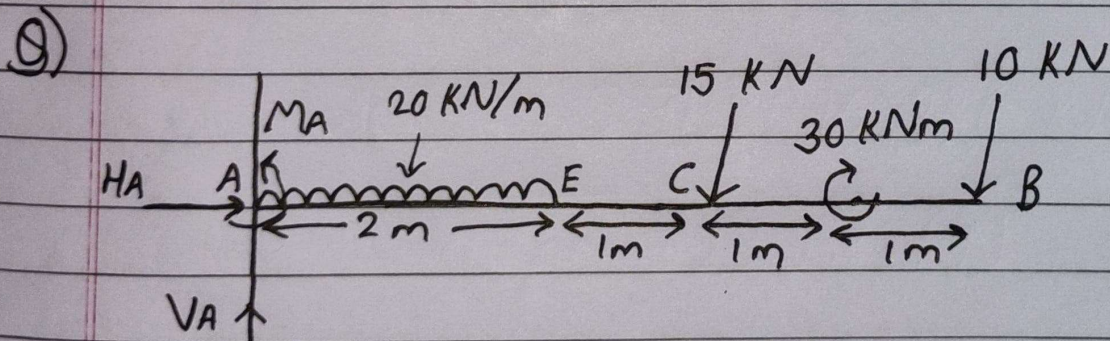
Solⁿ:



$$H_A = 0$$

$$V_A = 32 + 20 + 12 + 10 = 74 \text{ kN}$$

$$M_A = 32 \times 1 + 20 \times 2 + 12 \times 3 + 10 \times 4 = 148 \text{ kNm}$$



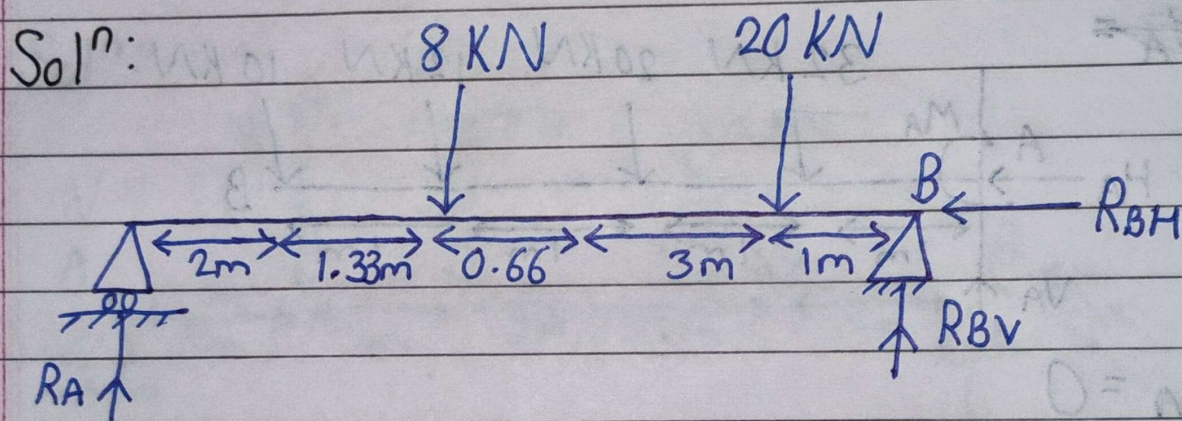
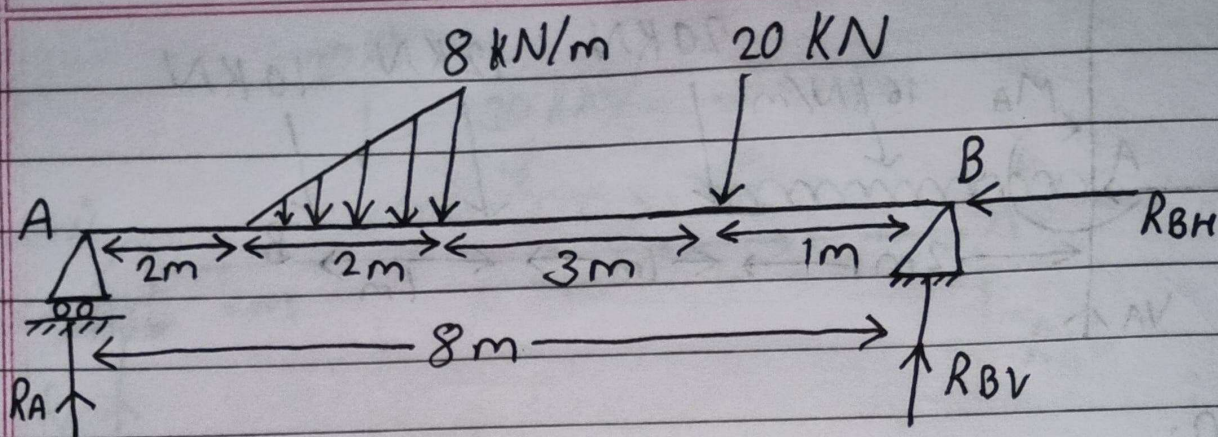
Solⁿ:

$$H_A = 0$$

$$V_A = 40 + 15 + 10 = 65 \text{ kN}$$

$$M_A = 40 \times 1 + 15 \times 3 + 30 + 10 \times 5 = 165 \text{ kNm}$$

9)



$$R_{BH} = 0$$

$$R_A + R_{BV} = 8 + 20 = 28$$

$$\sum M_A = 0$$

$$\Rightarrow 8 \times 3.33 + 20 \times 6.99 = R_{BV} \times 8$$

$$\Rightarrow R_{BV} = 20.805 \text{ kN}$$

$$\therefore \cancel{R_A = 8.805 \text{ kN}} \quad \therefore R_A = 7.195 \text{ kN}$$

$$\& R_B = 20.805 \text{ kN}$$